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Research on power plant security issues monitoring and fault detection using attention based LSTM model

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Abstract

Overview For Photo Voltaic (PV) arrays and Wind systems to operate as efficiently and effectively as possible, fault detection is essential. It is possible to improve the safety of renewable energy systems and guarantee that the service will continue uninterrupted if problem detection and diagnostics are performed in a timely and accurate manner. In general, wind power is one of the three major renewable energy sources, along with solar power and hydropower. Wind power is well distributed around the world, making it suitable to be exploited in human activities for the general welfare of society.

Objectives A prototype security situational awareness system applicable to the power data communication network service and traffic model should be developed. This will help to successfully enhance the security and service quality of the power data communication network, effectively cope with network security threats in the new environment, and ensure the security of the power plant network access. The traffic of the main network of the existing data communication network will be combined and analyzed, and NQA traffic management algorithms will be studied and proposed. These actions will improve the SLA hierarchical service capability, the service quality of the core services carried by the backbone network, and strengthen the security capability of the new energy power plant communication network access system.

Methodology For the purpose of this investigation, an attention-based long short-term memory (Att-LSTM) model was used for the categorization of time series actual data. The approach that has been developed is able to identify defects in photovoltaic arrays and inverters, which offers a dependable option for improving the efficiency and dependability of solar energy systems. For the purpose of evaluating the proposed method, a real-world solar energy dataset is used.

Results The findings acquired from this evaluation are compared to the results received from existing detection approaches such as Cryptography, Intrusion Detection System (IDS) methods, and Network Defense Schemes. The results obtained demonstrate that the suggested method surpasses current fault detection techniques, providing greater accuracy and better performance.

Keywords PV, Wind, Security monitoring system, NQA test instance, Hybrid system, LLLG fault, Attention based LSTM

Introduction

Upholding a consistent and long-term power supply is a major obstacle for the majority of emerging nations. The growth of most states' economies is stunted because of this. Numerous efforts have been made to produce power from renewable sources in an effort to mitigate the effects of energy shortages. The integration of renewable external sources with the current power system grid does present some technological obstacles, but [1]. Smart microgrids, which are the building blocks of the next-generation power grid, are cyber-physical energy systems. Investigating novel methods for physical fault diagnosis along with fault-tolerant control and intrusion detection as well as attack-resilient control will help smart microgrids become more reliable, secure, and cost-effective to operate [2]. Distribution and transmission line faults reduce the efficiency of the power grid, which in turn causes power outages and other disruptions in service. Enhanced economy and power supply dependability were the outcomes of faster fault diagnosis and accurate fault location estimation, which facilitated faster power supply reconstruction and maintenance [3].

Wind, solar, and battery units are among the DERs that are quickly making their way into the distribution power system. Because of the direct effect on power system operations, ensuring the cyber and physical security of these units is of the utmost importance in light of the efficiency and dependability of the power grid [4]. Renewable energy power production technology is becoming more mature, which is drawing more attention to distributed power generating based on renewable energy sources. Nevertheless, operational hazards to power supply security and reliability are heightened when DG penetration raises fault levels in the power distribution network [5]. Wind and solar power production has been a well-recognized alternative to conventional energy sources since the mid-nineteenth century. Still, power suppliers, operators, and consumers face additional obstacles due to the dynamic topology of distribution lines [6]. Because it is safe for the environment, solar PV technology is replacing traditional power generating methods in every country. Consequently, the current uses and applications of PV energy systems are driving the technology's rapid growth. PV arrays are powerful components of power systems, yet they are nevertheless vulnerable to certain failures. Economic returns, personnel and equipment security, accurate fault identification and diagnosis, and interruption mechanisms are all necessary for a successful supply [7].

Motivation

It is possible to improve the safety of renewable energy systems and guarantee that the service will continue uninterrupted if problem detection and diagnostics are performed in a timely and accurate manner. Several distinct approaches to fault identification and treatment for hybrid wind-solar power plants are going to be presented in this study [16, 17]. Our goal is to showcase these approaches. It is possible to divide the faults that are present in solar photovoltaic (PV) systems into two categories: transitory faults and permanent faults. Therefore, the transitory flaws are associated with meteorological conditions, such as the presence of clouds, partial shadowing, and the deposition of dust. When dealing with permanent problems, it is possible to conduct an investigation that takes into account early faults, extrinsic flaws, and degradation faults. On the other hand, the faults that are found in wind turbine power systems may be broken down into two primary categories: faults that are associated with the electric parts and faults that

are associated with the mechanical components. On the one hand, the defects that are associated with the electrical components are those that affect the performance of an electrical component of a wind turbine, for instance, the power converter/generator. On the other hand, the problems that are associated with mechanical components are those that affect a particular mechanical item, like the gear system, or the main bearings. All of the faults that were examined in this review have an impact on the primary components of the operating condition in solar PV systems and wind turbine generation systems. These faults have the potential to reduce the power output capacity of these systems, interrupt maintenance actions, and increase the costs of operation at the same time. For this reason, several approaches for performing fault identification and diagnosis have been documented in the literature. These methodologies complement one another and offer a solution to the problems that have been identified. It is extremely interesting to note that the direction of the investigations is towards the usage of data-driven strategies that use machine learning and deep learning approaches. This is the case in both SPV and WP. Acquiring data, pre-processing data, extracting features, reducing feature dimensionality, and classifying data are the five key processes that are involved in the majority of current systems for defect identification and treatment. It is important to note that these steps are generally referred to as the “5 fundamental steps.”

At its core, the suggested approach is to be described as.

- To begin, gather the sun and wind datasets; next, pre-process them to ignore or discard any irrelevant data.
- Then, with the aid of an Attention-based LSTM Model, the pre-processed structure may yield to the suggested model in order to discover the fault detection.
- In order to evaluate performance, the anticipated results, such as MSE, RMSE, and R2, were computed.

The Att-LSTM network is used as the residue generation in the approach. In order to forecast the output of a wind turbine benchmark, the enhanced network Att-LSTM learns all of the gathered variables, and then uses the residuals to determine the discrepancies between the anticipated and actual values. To further categorize the benchmark-defined defects, it integrates real-time signal processing with LSTM networks. When compared to the other three techniques on the benchmark, this method's simulation results demonstrate superior performance.

This is the remaining structure of the paper: In Sect. [Related work](#), the relevant research is described. In Sect. [Methods and materials](#), the Att-LSTM network structure and security difficulties are introduced. Section [Results & discussion](#) is devoted to tests and comparisons. Finally, in Sect. [Conclusion](#), the research effort is summarized and some future problems that require solutions are proposed.

Related work

The ground fault detection approach is shown in [8] for transmission line systems that are linked to inverter-based resources, including PV systems, electronically coupled wind turbines, and the like. When power system failures occur, IBRs, in contrast to traditional synchronous generators, may exhibit an initially high-frequency transient response with restricted magnitudes of short circuiting current. Because they compromise the element's security or reliability, IBRs' fault contributions could affect distance

protection. When IBRs are the primary source of fault current, distance protection could malfunction and trip for reverse faults or fail to trip for forward faults. When a single-line-to-ground fault occurs, the suggested ground fault detection technique may pinpoint its exact position by analyzing the dc voltage change in the inverters' dc-link. By interacting with the inverters over low-bandwidth channels, the technique is intended to assist distance protection in tripping forward SLG faults more quickly and avoiding tripping for reversal faults.

A hybrid AC/DC microgrid that uses wind and solar electricity to reduce the amount of DC/AC/DC power conversion procedures is suggested in [9]. The volatility of MG is increased when renewable energy penetration rates are high. Keeping the amounts of production and consumption in check may help reduce this volatility. That research proposes a coordination control scheme to stabilize the system in the face of generational, load, and fault variability by coordinating the flow of power over AC and DC connections. The MG uses a synchronous reference frame digital proportional resonant current controller to operate a bidirectional main converter. The PR controller reduces harmonics by acting as a digital filter having infinite impulse response characteristics, thanks to its high gain at the resonant frequency. In addition, the PR controller that is used can easily adjust to variations in grid frequency, has no steady-state error, is fast to follow the reference signal, and is straightforward to set up.

A smart hybrid microgrid that uses renewable energy sources and various distributed generating units, such as solar PV arrays, wind turbines, with battery energy storage systems, is discussed in the article [10]. For microgrid-level pulse-width modulation converters, new active FTC and ARC techniques are developed. The suggested fault-tolerant controller takes into account the negative effects of PV power-loss problems by using an optimum fuzzy gain-scheduling method. In addition, when data integrity cyber-attacks occur, the suggested attack-resilient controller uses the predicted values of sensor readings. The FTC and ARC approaches use a fuzzy modeling and identification-based integrated model-based intrusion detection as well as fault diagnosis platform to monitor and assess the micro grid's current health state in real-time. An advanced microgrid benchmark is used in a series of MATLAB/Simulink simulations to prove that the solutions are successful.

Explores the ever-changing power distribution behavior in the face of cyber, physical, and natural disasters as it pertains to reactive power regulation in DER units in [11]. Presented below are a number of possible fault scenarios, each of which involves a change in the pattern of the voltage-reactive power (Volt-Var) curve—its slope, dead band, and so on—as well as a shift in the reactive power supplied to the grid by capacitive or inductive means. In addition, they look at potential failure situations where dynamic VAR (volt-ampere reactive) compensating devices are included. They use Matlab/Simulink and ETAP, two popular simulation systems, to examine failures in a 140-bus commercial distribution electrical systems in southern California. According to the findings, changes to the reactive power control levels at DER units have a major impact on the system's voltage quality and security, as well as the frequency of VR switching operations.

Presented a new approach to fault detection in renewable microgrids using machine learning methods in the context of decentralized electrical systems in [12]. That research looks at how well machine learning models can spot irregularities in microgrid settings that are both dynamic and unpredictable. Energy storage status, fault indicators, and

metrics for solar, wind, or hydro power production are all part of the extensive information that it uses. The analysis shows that machine learning is better to traditional rule-based systems, which only managed an accuracy rate of 80%, with a remarkable 94% precision in defect identification. Measures of accuracy and recall highlight the machine learning models' balanced performance, which guarantees accurate issue identification while decreasing false positives and negatives. The models' capacity to sustain an effective energy supply is shown by the fact that the impact of failures on microgrid effectiveness is substantially mitigated, despite the only 2% drop seen in fault scenarios.

As it is intended to be linked to the city of Kufra, a solar generating system in Al-Kufra, Libya, is studied and analyzed in [13]. The ATP/EMTP software models and simulates a PV system, producing visual results. The first step is to create a virtual environment that mimics typical system functioning, complete with typical radiation levels and ambient temperature. Furthermore, by raising the ambient temperature to an appropriate level using proving radiation, the ensuing parameters will be examined and contrasted in every instance. Secondly, they will simulate and examine the effects of symmetrical and unsymmetrical failures in various power transmission lines on other electrical network characteristics.

In [14], the authors take into account various demands, use the correlation depth wandering method to evaluate distribution network node faults, suggest a quantitative evaluation approach for the capacity to regulate resources on the demand side, build an index system to evaluate the distribution network's resource regulation capabilities, and finally, evaluate the capacity to regulate resources on the demand side. Wind, thermal, solar, and hydropower all have regulatory efficiencies over 20%, meaning that when combined, these four energy sources may have efficiencies of over 80%. The study's findings strongly support the electric power industry's sustainable development and have important practical implications for resource regulation and security in distribution networks. The results demonstrate that the capability of multi-demand-side resource regulation for network security can be greatly enhanced.

There will be a steady march toward digitalizing security systems across all industries under the framework of the "dual carbon" national policy [15]. Research on the distribution grid's and emergency response systems' capacity to securely store and transmit renewable energy sources like wind and solar is acknowledged as crucial, as it forms the backbone of the distributed new energy sector under the dual carbon policy. The introduction of numerous energy storage plants using two-way power flow through the distribution network, along with the unpredictable and intermittent nature of photovoltaic wind power, causes a shift in the distribution network's characteristics from traditional power flow to distributed power flow. That, in turn, raises the load's complexity and variability and poses risks to the network's safe operation. Consequently, in order to manage the influence of the aforementioned additional energy on the power grid, energy storage devices are often used. In order to guarantee the grid's security with a large percentage of renewable energy, the energy storage facility may absorb energy, modulate frequencies, and reduce peak loads. Hence, new types of energy storage facilities are developed according to the needs of the times.

Methods and materials

Research gaps

The information state under normal conditions and the data state under fault or abnormal conditions of the wind-solar complementing power generating system are quite similar [18, 19]. In order to further differentiate between normal and aberrant data, it is recommended to use differential and fusion analysis on the collected data. Inadequate scalability, numerous dead zones, and limited monitoring coverage are just a few of the problems plaguing the current security surveillance system for wind-solar complementary power generation. What's more, the system fails to accurately detect and track anomalies and faults in this type of power generation, making it unsuitable for the rapidly expanding field of wind-solar complementary power generation. As a result, we built an IoT-based security monitoring system to keep an eye on our wind-solar complementary power generating projects. The prediction effects of deep learning algorithms have proven outstanding, leading to their widespread application in a variety of contexts. An effective deep learning model for time series data prediction is the long short-term memory neural network (LSTM). But now that data gathering is easier than ever thanks to technological advancements, multivariate time-series information are a thing of the past. There is usually a lot of data, a narrow timeframe, and a lot of connected sequences in multivariate time series data. The rules for changing numerous sequences will be impacted by the modifications made to other sequences, particularly in real-world data sets. When faced with this kind of data, deep learning models' prediction accuracy is severely compromised by interacting factor data, mutation data, and other problems. In contrast, we may concurrently extract information about the mutual effect of distinct sequences and utilize it as input to the model to improve the accuracy of the predictions.

Network security system awareness components

- (i) **Solar-Wind Hybrid System:** This system uses two power sources:
 - a. **Solar and Wind power:** The term "Solar-Wind Hybrid System" describes a power generation setup that makes use of both solar and wind power. Solar panels absorb light from the sun and transform it into electricity. Using wind turbines, the kinetic energy of the wind is transformed into electric energy by means of the wind's application to the blades. A more effective utilization of renewable energy sources is achieved by the combination of the powers produced by this hybrid system.
 - b. **Grid Supply:** In the event that renewable energy sources like solar and wind are unable to meet the system's power needs, the grid supplies is used for the backup system.
- (ii) **Energy Meter:** To find out how much power is being used from the grid, one uses an energy meter. The grid supply is employed as a backup system, with the Smart System installed prior to the load, in order to manage or save energy in a more intelligent manner.
- (iii) **Fault detector:** Using a deep learning model based on a time series method, this module diagnoses wind and solar PV information to defect identification.

Network security situation assessment and prediction

The ability to foretell, estimate, and analyze potential future changes in the state of network operations based on current and historical data, scientific theories, methods, and a variety of experiences, judgments, and knowledge is a crucial component of network security awareness of situations. By evaluating and building the decision matrix, the system concludes the assessment of the network security situation, determines the weight of indicators, and extracts features of the network situation using an analytical hierarchy method. Foresight into security policy setup, dynamic network security leadership, and the prevention of massive threat network security incidents are achieved by using the evaluation data to inform the selection of appropriate network security situation forecasting technologies. Figure 1.

By integrating data, analyzing it, and visualizing it, among other technologies, network situational awareness provides security managers with a dynamic and intuitive view of the operating status as well as security situation of information systems. This allows

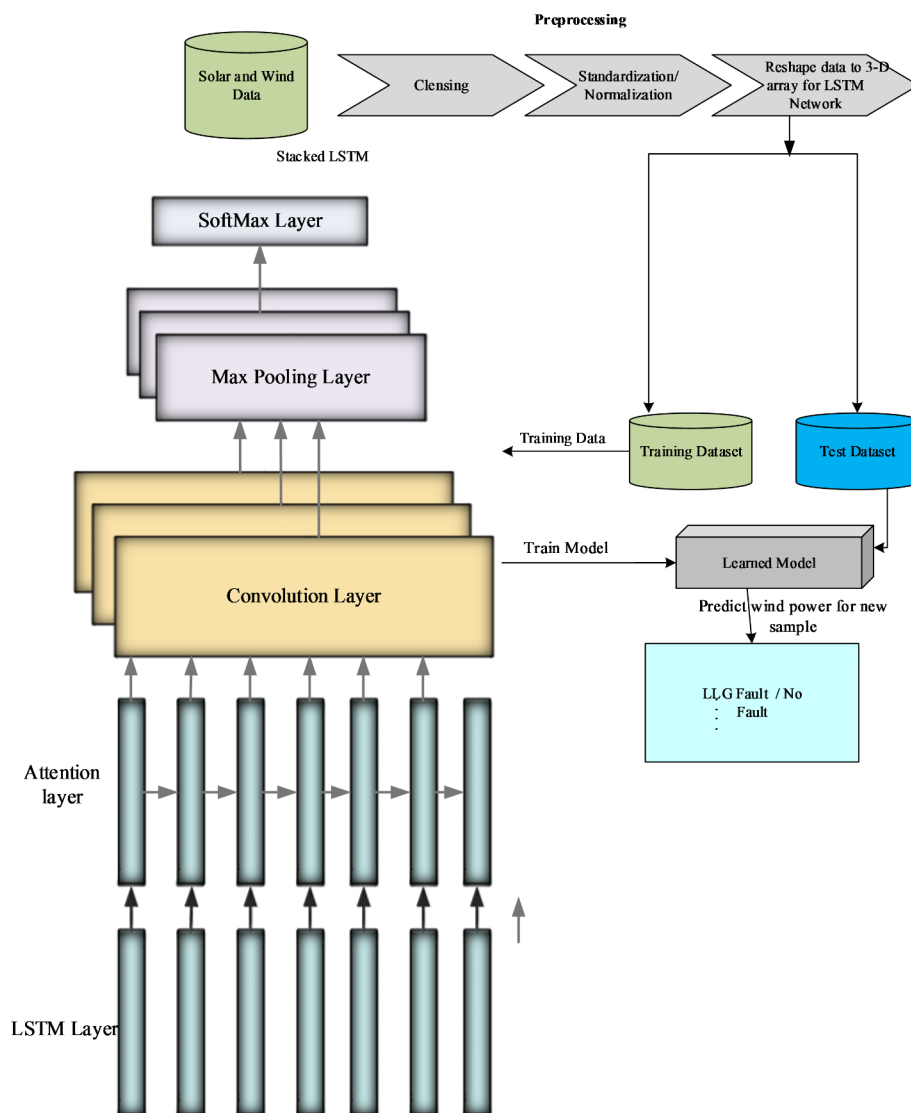


Fig. 1 Att-LSTM Model for Fault Detection

them to see current network security position in real time and to anticipate information risks. So, to summarize, the perception platform's primary features are:

- Integration of data: In order to effectively assess the network security status, the awareness platform should normalize fusion of data from various network devices, which will increase data transmission speed and offer comprehensive sources of information for situational awareness analysis.
- As a monitoring and prior warning platform, its primary application is to facilitate the data analysis level's large data storage, computation, and query functions. Information pertaining to the state of network security may be found in security logs as well as in traffic from outside the network. The platform primarily gathers data from the aforementioned sources and keeps an eye on the website. The data retrieval feature of the hot node tracking platform is used, which includes access times and traffic, and data is saved, computed, and queried via the application cluster and dispersed log. On top of that, the information network web crawler while website surveillance and alarm routinely carry out activities including regulating and seizing the attacker's account as well as accessing desired resources.
- Perceive attack risk: situational awareness is primarily concerned with network attacks and threats. Attacking behavior is analyzed to determine the signs of a threat, and all relevant information, including threat modules, is combined with modeling. Once this information is found, it can be used to check for threats and achieve early warning functions for potential attacks.
- Data visualization: managers can intuitively view the growth trend of ticket information as well as quickly detect security vulnerabilities by transforming a large quantity of database data into a trend map as well as distribution map via an array of connection and combination techniques. Concurrently, various user demands should be taken into account when generating targeted situation assessment reports. These reports should serve as auxiliary functions for efficient processing and subsequent decision-making [20].

Data acquisition

Here are some key points about the data acquisition process for the wind-solar power generation system: the wind speed mistakes are all less than 0.6 m/s, and the temperature data acquisition errors are all smaller than 0.5 °C. Additionally, the data acquisition is done three times, and all of the data are identical to the actual value.

The Nanshan Power Plant in Hunan, China, has been running for nine years and features ten wind turbines and solar photovoltaics. Every one of the wind turbines is a 2 MW model. Solar photovoltaic (PV) and wind turbine primary components are equipped with sensors by the SCADA system. Whenever the wind turbine is in operation, we take note of its operational parameters and save them for later use. Different units and storage formats are also used for these characteristics. The number of variables is close to one hundred. A few examples of the characteristics that are relevant to the pitch system are the following: wind speed and direction, hub speed, pitch angle, acceleration of vibrations, generator power output, and the like. It is obtained from the SCADA system of the wind turbine.

There are two components to the input data: data produced by the plant and weather data collected at the plant. Columns containing input data from these two databases include.

- Date time.
- Irradiation at the solar plant.
- Wind Speed.
- Yield of the plant.
- Solar PV Temperature.

The system was subjected to inrush current fluctuations from transformers and capacitors as well as switching load and loads for 0.066 s to 0.15 s in order to analyze the non-faulty event. For the analysis, a 400 kVar to 500 kVar capacitor switching and a 0.5 MW to 2 MW load switching were employed, with each step of 50 kVar. For example, from 0.066s to 0.15s, a 1 MW load switches 300 kV capacitors.

Remove the 2600–4400 s timeframe from the acquired data, and choose a point every 0.1s to prevent very frequent diagnosis. Following that, the data is partitioned into three sets, each with fifteen variables: the prediction set (2600–2800 s), the train set (2800–3600 s), and the examination set (3600–4400s). With Butterworth low-pass filters, these will be filtered.

Pre-processing of input data

A labeled time series dataset is written as, $D = \{(x_i, y_i)\}_{i=1}^T$. Then we have $x_i = (x_i^1, x_i^2, \dots, x_i^n)^T \in \mathbb{R}^n$, which signifies a vector of n exogenous (driving) input time series at the i^{th} time step, and $y_i \in \{1, \dots, C\}$ by the by $C \in \mathbb{Z}^+$ is the total number of labels.

We extract the values mentioned previously. Prior to transmitting the data for prediction, the following aspects are verified.

Verify that no values are missing

Combine the databases by syncing the input dates

Further, some other processes are used for fault detection as follows, and then we proceed to preprocess the dataset. Following is a list of the primary steps involved in preprocessing:

- **One-hot Encoding:** For our deep learning model to produce the best possible prediction results, we need to convert the categorical characteristics found in the dataset into numerical values. For this reason, the columns were converted into numerical values. This was done during the preparation step. We opted for one-hot encoding over the label encoder because the latter will generate several numbers in a single column, which could lead the model to incorrectly assume a certain order for the data, impacting categorization;
- **Normalization:** The process of rescaling data to a particular range to avoid redundancy and speed model training. Normalization is often referred to as “regulation.” A min-max normalization is utilized in the research project to resize the data range to [0,1]:

$$x[i] = \frac{x[i] - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

- **S-fold Stratified Cross-Validation:** The process of stratification entails restructuring the data to ensure that each subset is an accurate representation of the complete dataset. Stratified S-fold cross-validation involves splitting the dataset into S subsets, with each subset serving as a training ground for the model. The model is then validated on this final subset. This procedure will continue until all of the folds have been investigated and used to validate the model. The strategy of stratification ensures that each fold accurately represents the entire dataset, which makes parameter optimization easier and improves the model's capacity to classify attacks.

Fault prediction

Based on the data shown above, it can be concluded that the proposed system acquires data on the wind-solar complementing power generation network with little mistake and high dependability. This data provides a solid foundation for tracking the security of the system. The suggested system will first collect data from the typical wind-solar CPS system, and then it will analyze the program interruption phenomenon to make sure the data is intact, lessen the CPS system's data error, and give a solid foundation for monitoring the CPS system [21, 22].

In order to determine the important timeframe for learning, this research used an Attention-based Long Short-Term Memory (A-LSTM) model over several forecasting time dimensions.

LSTM's ability to selectively remember or discard incoming data makes it more effective than regular RNN. On the one hand, it possesses a powerful memory block that primarily consists of 3 types of gates such as forgetting gate, memory gate, and output gate. On the other hand, it also possesses a memory unit that can govern the transfer of information to the subsequent moment. The essential components of an LSTM unit structure are the unit state, input gate, forget gate, and output gate.

The input is denoted by Z , while the input gate's control signal is denoted by Z_i . The forget gate's control signal is denoted by Z_f , The output gate's control signal is denoted by Z_o , and the Sigmoid function is often taken as the $f(x)$ function:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

The value range for this function is between 0 and 1, and it is used to indicate the degree to which the door is opened from opening to closed. The activation functions $g(x)$ and $h(x)$ are both the same thing. To begin, Z is passed the activation function to obtain $g(Z)$. Next, Z is passed the Sigmoid function to obtain $f(Z_i)$ and finally, Z is multiplied to obtain $g(Z)f(Z_i)$. Last but not least, Z_f multiplies $f(Z_f)$ by the value of the memory unit a to get $cf(Z_f)$ which is obtained by passing it through the Sigmoid function. In conclusion, the memory unit value should be updated to:

$$c' = g(Z)f(Z_i) + cf(Z_f) \quad (3)$$

The activation function is passed to c' to obtain $h(c')$, and Z_o is passed to the Sigmoid function to obtain $f(Z_o)$. The output is then multiplied to obtain the following:

$$a = h(c')f(Z_o) \quad (4)$$

At time t , diverse weight vectors multiply the hidden layer neuron's output h_{t-1} at time $t-1$ with the input x_t at time t , creating a combined input to the hidden layer. To acquire the control signals of the three gates Z_f, Z_i, Z_o , as well as the input value Z , utilization of the activation function is required. A formula can be written as follows:

$$\begin{aligned} Z_f &= \omega_f \cdot [h_{t-1}, x_t] + b_f \\ Z_i &= \omega_i \cdot [h_{t-1}, x_t] + b_i \\ Z_o &= \omega_o \cdot [h_{t-1}, x_t] + b_o \\ Z &= \omega_x \cdot [h_{t-1}, x_t] + b_x \end{aligned} \quad (5-8)$$

A few examples of these biases include b_f, b_i, b_o , and b_x which symbolize various connection weights. Following the operation of the LSTM unit, the value c of the memory unit is revised in Eq. (2), and Formula (9) is achieved. The below formula represents the output of neurons in the buried layer:

$$\begin{aligned} c' &= g(\omega_x \times [h_{t-1}, x_t] + b_x) f(\omega_i \times [h_{t-1}, x_t] + b_i) \\ &\quad + cf(\omega_f \times [h_{t-1}, x_t] + b_f) \\ h_t &= h(c') f(Z_o) \\ &= h(g(\omega_x \cdot [h_{t-1}, x_t] + b_x) f(\omega_i \cdot [h_{t-1}, x_t] + b_i) \\ &\quad + cf(\omega_f \cdot [h_{t-1}, x_t] + b_f)) f(\omega_o \cdot [h_{t-1}, x_t] + b_o) \end{aligned} \quad (9\&10)$$

From the equations presented above, it is clear that the input of the LSTM not only incorporates the output h_{t-1} of the hidden layer neuron at the most recent instant, but it also incorporates the value of the memory unit that is included within the LSTM unit themselves. In addition to being able to recall long-term historical information and fit long-term time series data more efficiently, the LSTM network is also capable of successfully avoiding the occurrence of gradient disappearance.

Two different types of neural networks make up LSTM: an attention-based LSTM neural network and a convolution neural network. When we get to the following sections, we will go into detail on each of the components.

Some ways to categorize the attention process include spatial attention and temporal attention, which are based on the relative importance of different locations and times, and soft attention and complex attention, which are based on the specifics of the task at hand. Multiple or even dozens of related sequences are included in multivariate time series data. Any changes that occur in any of these sequences will affect the target sequence. The attention mechanism determines the attention value of every sequence by calculating the distribution weight of every sequence based on past lows. This measure is the attention value of every sequence. Finally, depending on the attention value of every sequence, do an analysis and make a prediction.

The LSTM unit is iteratively computed using the following input time series: $x_1, x_2, \dots, x_T$, where $x_1 \in \mathbb{R}^n$.

$$h_i = \sigma(W[h_{i-1}; x_i] + b_s) \quad (11)$$

The concatenation of the prior hidden state h_{i-1} and the input x_i at time step i is represented by the expression $[h_{i-1}; x_i] \in \mathbb{R}^{m+n}$. The weight matrix, denoted by $W \in \mathbb{R}^{m \times (m+n)}$ is a weight matrix. Following that, the weighted feature of time step i is computed based on the attention mechanism in the following manner:

$$l_i = \sum_{j=i-T_x}^i \alpha_j h_j \quad (12)$$

The computation of the attention weight α_j is performed in the following manner, where T_x represents the windows size of the attention weight computation:

$$\alpha_j = \frac{\exp(h_j^T q_s)}{\sum_{k=i-T_x}^i \exp(h_k^T q_s)} \quad (13)$$

During training, the vector q_s which is initialized at random and utilized to choose the most applicable hidden states over all time steps, will be optimized.

Convolution Neural Network: One-dimensional convolutional layers and one-dimensional max pooling layers make up the Convolution Neural Network and its components. To obtain the feature $c_i \in \mathbb{R}^s$ of time step i , we first input the time-series data $x_{i-T_x:i}$ into CNNs Network. These data are then processed.

Classification: Utilizing a weight matrix $U \in \mathbb{R}^{C \times (m+r)}$ and bias $c \in \mathbb{R}^c$, the prediction label of the time series is obtained by employing a softmax layer.

$$y_i = \text{softmax}(U[l_i; c_i] + c) \quad (14)$$

The features from the attention-based LSTM network and the CNNs network are combined to form $[l_i; c_i] \in \mathbb{R}^{m+s}$.

Results & discussion

The experimental data shows the wind speed and temperature collected by the suggested system, a security monitoring system based on MATLAB.

By utilizing the MATLAB function `predictAndUpdateState`, we are able to ascertain the future values of several time steps. Each prediction updates the network state, and the process continues in this way until the values are predicted at each time step.

The experimental assessment that was carried out to verify and assess the claims made in the study is explained in this section. Additionally, a comprehensive explanation of the experimental setup is included here. After that, we will conduct a comprehensive analysis of our findings and results. PV systems are susceptible to a wide variety of abnormalities. Tests were carried out to study the impact caused by both internal and external factors, as well as the correlation effect, on the data of all inverter sensors for both plants.

This was done so that an accurate comparison could be made between the anomaly detection algorithms that were being utilized. The many benefits of PV power generation—including being pollution-free, safe, noiseless, easily installed, and requiring little in the way of building time—have led to its increasing use. Solar cell modules should last for about 20 years in principle, but in reality, some of them break down after just 8 to 10 years of operation for a variety of reasons. For this reason, a PV power station monitoring system that can identify and fix problems quickly is crucial for the safe and dependable operation of these facilities. Deterioration owing to age or operating occurrences, such as the membrane drying out or becoming dehydrated, fuel/gas of the electrochemical reaction (as a result of variations in channel flow or flooding), and membrane leakage are the major types of faults. A voltage drop is the usual result of all fault situations. Solar radiation, the temperature, electricity, and voltage production are some of the operational parameters that should be included in this dataset. Labeled data showing the existence or absence of defects, as determined by human inspection or monitoring systems, should also be included. Extracting useful characteristics from a dataset is

the next step after collection. The measured data may be used to calculate properties like the MPP, voltage-current characteristics, and temperature coefficients, among others. With the use of feature extraction, we may extract the most relevant information for defect identification while also reducing the data's dimensionality. Optimization of the network's intrinsic measurements, usually via the use of backpropagation underlying gradient descent methods, allows it to understand the data's underlying patterns and correlations. Minimizing the discrepancy between the actual fault labels and the network's anticipated outputs is the objective. Solar power facilities utilize a variety of sensors to monitor their performance and efficiency. The authors used inverter, temperature, irradiance, and current sensors in their study. Only 96 out of a total of 100 data points are actually used for training purposes. On overcast days, the writers culled four pieces of data. In order to evaluate different diagnostic methods, we present a list of ten qualities an ideal diagnostic system must possess:

- Rapid detection and detection: the system should react rapidly to identify and diagnose process malfunctions.
- Isolability: the capacity to differentiate between various faults.
- Novelty identification: the diagnostic system's capacity to differentiate between new, unknown defects and other known malfunctions or normal operation.
- Robustness: a diagnostic system's capacity to withstand different types of noise and uncertainty.
- Classification error estimate: a diagnostic system's capacity to predict, in advance, the likelihood of a classification error occurring; adaptability: a diagnostic system's capacity to adjust to new circumstances, such as fluctuating production quantities in response to fluctuating demands or changes in the quality of raw materials; explanation facility: a testing system's capacity to explain where the fault came from and how it spread to the current state of affairs; modeling demands: a system for diagnostic purposes should have minimal modeling effort for efficient deployment; storage and computational aspects demands: the ability to strike an acceptable compromise between computational demands and needs for storage.
- The capacity to detect several problems simultaneously, which is challenging since most errors interact with one another.

Performance analysis

For this model's assessment, we relied on Mean Absolute Error (MAE) with Root Mean Square Error (RMSE). Concurrently, while comparing our model to others, we used MAE and RMSE as our assessment markers.

Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were the indicators that we utilized to quantify the performance of the model. The mean absolute error (MAE) and root mean square error (RMSE) were also utilized as indicators of assessment to compare experiments with different models. These two assessment indicators each have their unique calculation algorithms, which are as follows:

$$\begin{aligned} MAE &= \frac{1}{N} \sum_{i=1}^N |y_i - x_i| \\ RMSE &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - x_i)^2} \end{aligned} \quad (15\&16)$$

Here, N is the test set's length, y_i is the value predicted by the model, and x_i is the value that occurred. The three indices also show that the suggested Att-LSTM is better than other data-driven methods, including IDS [13], Network Defense [15], and Security schemes [14]. On the assumption of the same input data, Att-LSTM performs better than earlier methods when dealing with time dependency. It is critical to employ the Att-LSTM network for real-time signal processing and classification and to fully use the residual signals' correlation of fault information with respect to time and location. This will let the network fully mine the process data sequence and variable correlations. For the above parameters, the comparison is implemented for the following works:

- Approaches to classification: In the absence of further information about the relationships between characteristics and defects, methods including statistical, neural, neurofuzzy, geometrical, fuzzy clustering, and support vector machines (SVMs) may be used. Figure 2.
- To isolate the source of the problem, inferential methods may be used if the linkages between the causes of the problem and its symptoms are recognized, or at least somewhat understood. Some possible representations of causal linkages include fault trees, fuzzy relations, and expert system knowledge.

Discussion

This research looks at renewable energy sources that are linked to the power grid and how they may be controlled and simulated in real-time. The only two types of systems that were considered for inclusion in our research were wind and PV systems that were connected to the power grid through the point of common connection (PCC). There are approaches for simulation and control that are presented. The dynamic performance of the hybrid system that has been suggested is evaluated under fault situations that involve the line-to-line line-to-ground fault (LLLG fault). The findings of the simulation demonstrate that it is possible to control the tiny hybrid system that combines wind

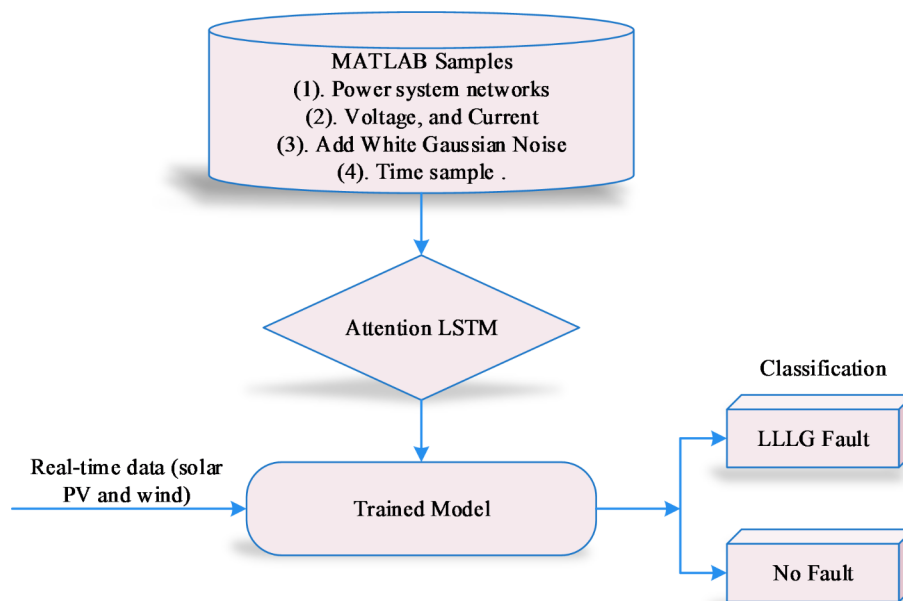


Fig. 2 Att-LSTM Model for Fault Classification

Table 1. The predicted results for hybrid modeling

	MAE	MAPE	RMSE	PCC	R2	NSE
<i>Solar PV Phase</i>						
Fuzzy Logic	0.043	35.267	0.072	0.967	0.044	0.914
Fuzzy Clustering	0.041	33.745	0.070	0.970	0.044	0.921
ANN	0.019	22.100	0.045	0.990	−58.588	0.973
Neuro Fuzzy	0.021	22.528	0.047	0.989	−69.997	0.971
SVM	0.000	29.597	0.000	1.000	0.000	1.000
Proposed	0.001	29.711	0.002	1.000	0.000	1.000
<i>Wind Turbine Phase</i>						
Methods						
Fuzzy Logic	0.031	6.876	0.049	0.989	17,460.222	0.989
Fuzzy Clustering	0.031	6.866	0.048	0.989	17,140.021	0.9
ANN	0.039	22.150	0.051	0.985	−58.638	0.968
Neuro Fuzzy	0.041	22.578	0.053	0.984	−70.047	0.966
SVM	0.000	0.083	0.000	1.000	0.000	1.000
Proposed	0.000	0.091	0.000	1.000	0.000	1.000

and solar electricity, as well as the dynamic performance of the system. In contrast to intrusion detection systems (IDS), neural network methods process temporal sequences using neural processing, allowing one to manage the input values to the present-time model while keeping their impact on the unit state gates. In addition to utilizing multiple parameters as attribute items for input data, this piece embraces the attention mechanism of various channels. Attributing relative importance to various characteristics is fundamental to the attention process. The most heavily weighted characteristic in wind turbine defect warning is the most essential, but all of the attributes' influences on the prediction results are kept intact and incorporated into the warning results, making the warning more effective overall. Table 1.

The findings show that when compared to other intelligent classifiers, the suggested deep learning approach of recurrent LSTM classifier detects more fast. A personal computer with these requirements was used to record the detection time results: 32-bit operating system, 8 GB of RAM, and x64-based CPU (Intel (R) core (TM) i5-2410 M CPU @2.30 GHz).

(1) Both the present condition and the operational wear of the past are connected to the fault characteristics of the pitch system. To better depict the device's operational condition, we utilize a sliding window to collect the time series of parameters. These parameters may then be supplied to LSTM. (2) In order to unify parameters of different dimensions into ratio vectors, we use parameter fluctuations to generate multi-parameter ratio vectors. By feeding the Att-LSTM with the many parameters ratio vector, we can make better predictions and make sure the model stays stable. (3) To enable the model to extract the signal's temporal properties, we utilize an attention mechanism to enhance the network's long-term and short-term memory. Then, we acquire fusion information at different scales through the interplay of cross-channel feature information.

Conclusion

When failure situations are considered, like the LLLG fault, the dynamic performance of the hybrid system that has been recommended is examined. As a result of the durability of the controller, the parameters of the system quickly recover to their initial levels when a failure has occurred. The simulation results validated the effectiveness of the

control methods employed, as the produced reactive and real powers align with the reference values supplied by the solar system's MPPT control and the wind turbine's pitch angle control. The assessment findings on a real-world solar energy dataset show that the suggested strategy achieves better performance and greater accuracy than current fault detection algorithms. In addition, the suggested method may be applied to many renewable energy systems, laying the groundwork for the creation of all-encompassing frameworks for defect identification and diagnosis. The areas of solar energy systems and problem diagnosis and detection have benefited greatly from this study. With the ever-increasing need for clean energy, the suggested method provides a viable answer to the problem of how to make renewable energy systems more reliable and efficient. More effective and long-lasting use of renewable energy resources is possible with the aid of the suggested method, which may extend the life of solar power systems and decrease their maintenance costs. More efficient and dependable renewable energy systems may be in the works as a result of this study's implications for defect detection and diagnostics in solar energy systems. In future, blockchain technology will be used for fault prevention to store the transactions in blocks. Further, faults preventive mechanisms will be further added to improve the performance of the power plants.

Author contributions

Han Xu and Ji Du wrote the main manuscript text and Shengda Wang, Zeng Dou and Danni Liu prepared figures. All authors reviewed the manuscript.

Funding

Supported by Science and Technology Projects(Research and Application of Security Protection Technologies for Network Access in new energy Power Plants)of Jilin Jineng Electric Power Communication Co., Ltd.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country.

Consent for publication

I confirm that any participants (or their guardians if unable to give informed consent, or next of kin, if deceased) who may be identifiable through the manuscript (such as a case report), have been given an opportunity to review the final manuscript and have provided written consent to publish.

Competing interests

The authors declare no competing interests.

Received: 12 November 2024 / Accepted: 18 January 2025

Published online: 27 January 2025

References

1. Kabouris J, Karystianos M, Nomikos BM, Tsourakis GI, Mantzaris JC, Sakellariadis NG (2022) Power system static and dynamic security studies for the 1st phase of Crete Island Interconnection. *Journal of Energy - Energija*
2. Kulshrestha A, Gupta MK, Mahela OP (2021) DISCRETE WAVELET TRANSFORM AND DECISION RULES SUPPORTED METHOD. FOR DETECTION OF FAULTY EVENTS IN DISTRIBUTION NETWORK WITH RENEWABLE ENERGY
3. Mekonnen HT (2021) Increment of Fault Current due to integration of distributed generation and its impact on distribution Network Protection. *Archives of Infectious Diseases & Therapy*
4. Lyu Z, Yang X, Zhang Y, Zhao J (2020) Bi-Level Optimal Strategy of Islanded Multi-Microgrid Systems Based on Optimal Power Flow and Consensus Algorithm. *Energies*
5. Gogula V, Edward B (2023) Fault detection in a distribution network using a combination of a discrete wavelet transform and a neural network's radial basis function algorithm to detect high-impedance faults. *Frontiers in Energy Research*
6. Sarangi RR, Ray PK, Barisal AK, Mohanty A (2023) Fault Detection in DC Nanogrid System using Advanced Signal Processing Techniques. 2023 11th National Power Electronics Conference (NPEC), 1–6
7. Moloi K, Hamam Y, Jordaan JA (2020) Fault Pattern Recognition in Power Distribution Integrated Network with Renewable Energy Source. 2020 5th International Conference on Renewable Energies for Developing Countries (REDEC), 1–6
8. Samkari HS (2023) Ground Fault Detection for Transmission Lines Connected to Photovoltaic Systems Using Inverters DC-Link Voltage. 2023 3rd International Conference on Computing and Information Technology (ICCIIT), 475–479

9. Jasim AM, Jasim BH, Neagu BC, Alhasnawi BN (2022) Coordination Control of a Hybrid AC/DC Smart Microgrid with Online Fault Detection, Diagnostics, and Localization Using Artificial Neural Networks. *Electronics*
10. Jadidi S, Badihi H, Zhang Y (2023) Active Fault-Tolerant and attack-resilient control for a renewable microgrid against power-loss faults and data Integrity attacks. *IEEE Trans Cybernetics* 54:2113–2128
11. Joseph A, Smedley KM, Mehraeen S (2021) Secure Power Distribution against Reactive Power Control Malfunction in DER Units. *IEEE Trans Power Delivery* 36:1552–1561
12. Dutt A, Rani MS, Bisht MS, Chandna M, Singla A (2024) Machine Learning Approaches for Fault Detection in Renewable Microgrids. *E3S Web of Conferences*
13. Saad SM, Sharif KI, Mohamed T, Mustafa HY, El Naili N, El-Arroudi K, Mohamed FA (2022) Modeling, Simulation, and Integration Considerations of Utility-Scale Solar PV to Assess FRT Detection-Libyan Sub-transmission Grid Case Study. 2022 International Conference on Engineering & MIS (ICEMIS), 1–5
14. Han Y, Lin X (2024) Multi-demand-side resource regulation capacity assessment for distribution network safety and security. *J Comput Methods Sci Eng* 24:2267–2282
15. Liu J, Ren Y, Dong Z, Yu Y, Xiong Z, Qiao L, Lu Y (2023) Research and Development of Monitoring and Early Warning Platform of Battery Energy Storage Power Station of New Power System. 2023 8th Asia Conference on Power and Electrical Engineering (ACPEE), 554–558
16. Said D, Mouftah HT A novel electric vehicles charging/discharging scheme with load management protocol, 2017IEEE International Conference on Communications (ICC), Paris, France, 2017, pp. 1–6. <https://doi.org/10.1109/ICC.2017.7997117>
17. Said D, Cherkaoui S, Khokhi L (2013) Advanced scheduling protocol for electric vehicle home charging with time-of-use pricing, IEEE
18. International Conference on Communications (ICC), Budapest H (2013) pp. 6272–6276. <https://doi.org/10.1109/ICC.2013.6655612>
19. Said D, Elloumi M (2022) A New False Data Injection Detection Protocol based Machine Learning for P2P Energy Transaction between CEVs, 2022 IEEE International Conference on Electrical Sciences and Technologies in Maghreb (CISTEM), Tunis, Tunisia, pp. 1–5. <https://doi.org/10.1109/CISTEM55808.2022.10044067>
20. Khabbouchi I, Said D, Oukaira A, Mellal I, Khokhi L (2023) Machine learning and game-theoretic model for advanced wind Energy Management Protocol (AWEMP). *Energies* 16(5):2179. <https://doi.org/10.3390/en16052179>
21. Alghamdi TG, Said D, Mouftah HT (2021) Prot Maximization for EVSEs-Based Renewable Energy Sources in Smart Cities With Dierent Arrival Rate Scenarios, in *IEEE Access*, vol. 9, pp. 58740–58754. <https://doi.org/10.1109/ACCESS.2021.3070485>
22. Alghamdi TG, Said D, Mouftah HT Decentralized Game-Theoretic Approach for D-EVSE based on Renewable Energy in Smart Cities, ICC 2020–2020 IEEE International Conference on Communications (ICC), Dublin, Ireland, 2020, pp. 1–6. <https://doi.org/10.1109/ICC40277.2020.9148718>

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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